

Data Visualization and Storytelling Report

Project: Netflix Movies and TV Shows: Data Visualization and Storytelling

Introduction

This project analyzes the Netflix Movies and TV Shows dataset using Python libraries including Pandas, Matplotlib, Seaborn, and WordCloud. The objective is to explore the dataset, create meaningful visualizations, and communicate insights through data storytelling.

Objectives

- Explore the dataset
- Clean and preprocess the data
- Create informative visualizations
- Identify trends and patterns
- Present insights effectively

Dataset

The dataset contains information such as title, type, director, cast, country, release year, rating, duration, genres, and description for Netflix movies and TV shows.

Tools Used

Python, Pandas, Matplotlib, Seaborn, WordCloud, Jupyter Notebook, Visual Studio Code.

Data Preprocessing

Missing values were handled, date columns were converted, categorical values were standardized, and movie durations were converted to numeric format where necessary.

Visualizations Created

1. Netflix Content Growth Over Time
2. Movies vs TV Shows
3. Top Producing Countries
4. Ratings Distribution
5. Genre Distribution
6. Movie Duration Distribution
7. Release Year Distribution
8. Description Word Cloud

Key Findings

Netflix experienced rapid growth after 2015. Movies dominate the platform, the United States produces the most content, Drama and International Movies are the most common genres, most titles target mature audiences, and typical movie durations range between 80 and 120 minutes.

Conclusion

The project demonstrates how effective visualizations can transform raw data into actionable insights and improve communication of analytical findings.

Future Scope

Interactive dashboards, recommendation systems, sentiment analysis, predictive analytics, and comparisons with other streaming platforms can further extend this project.